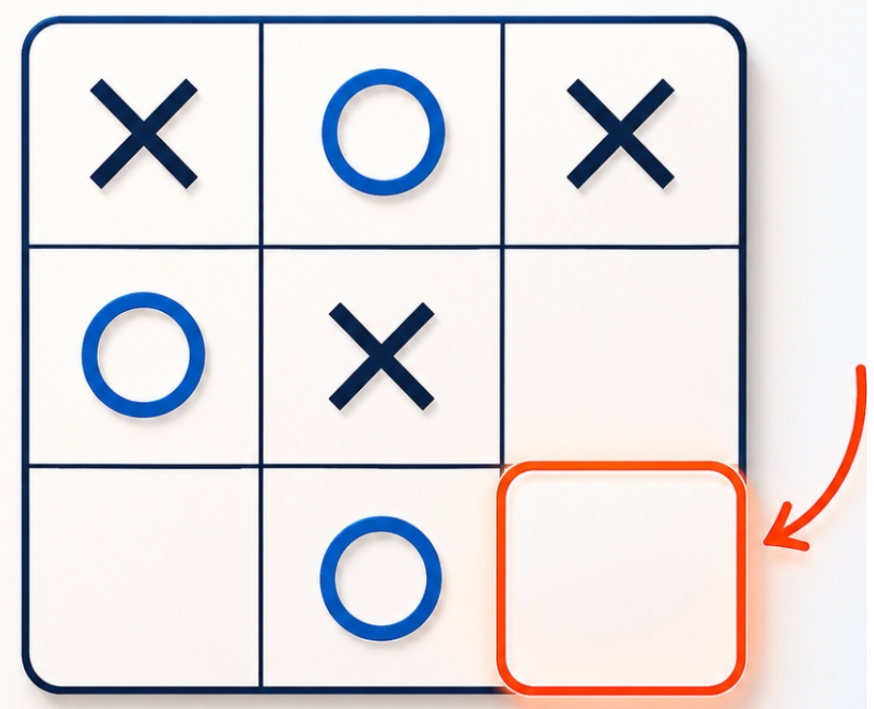
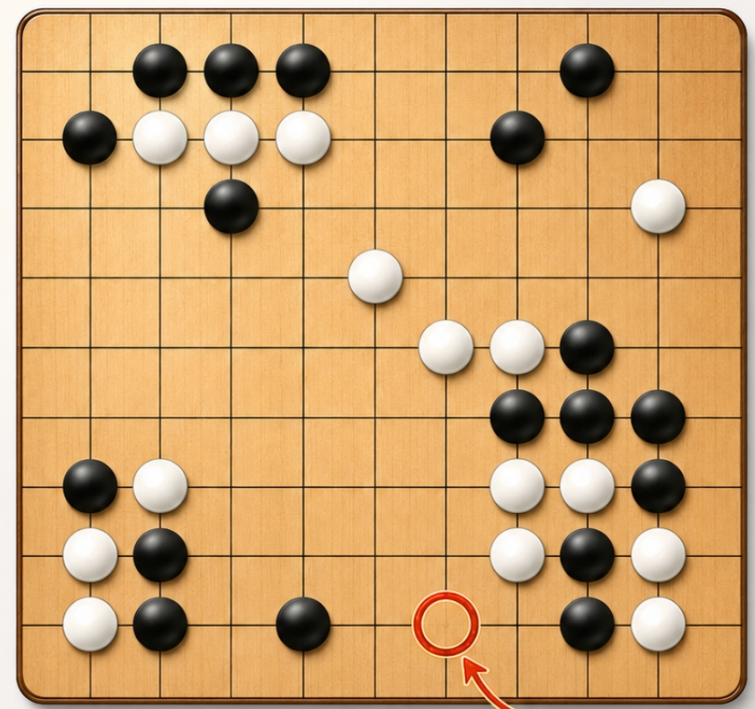
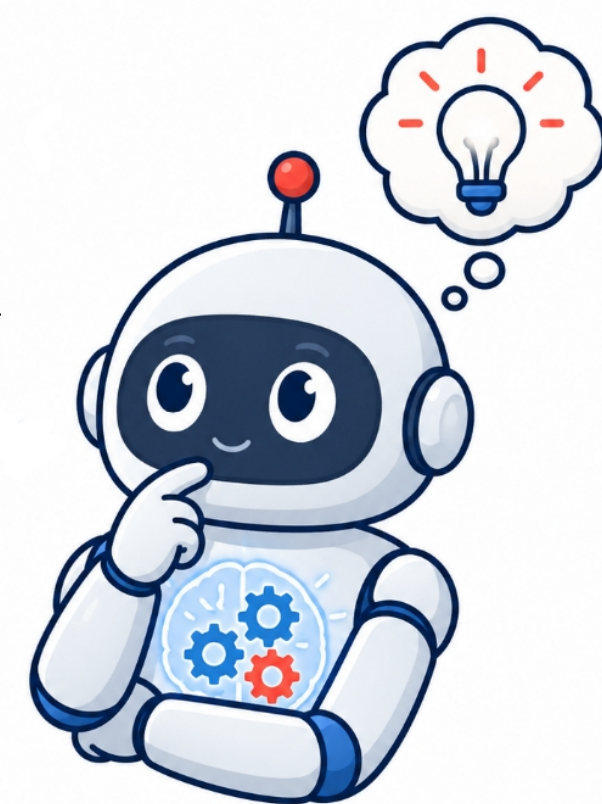
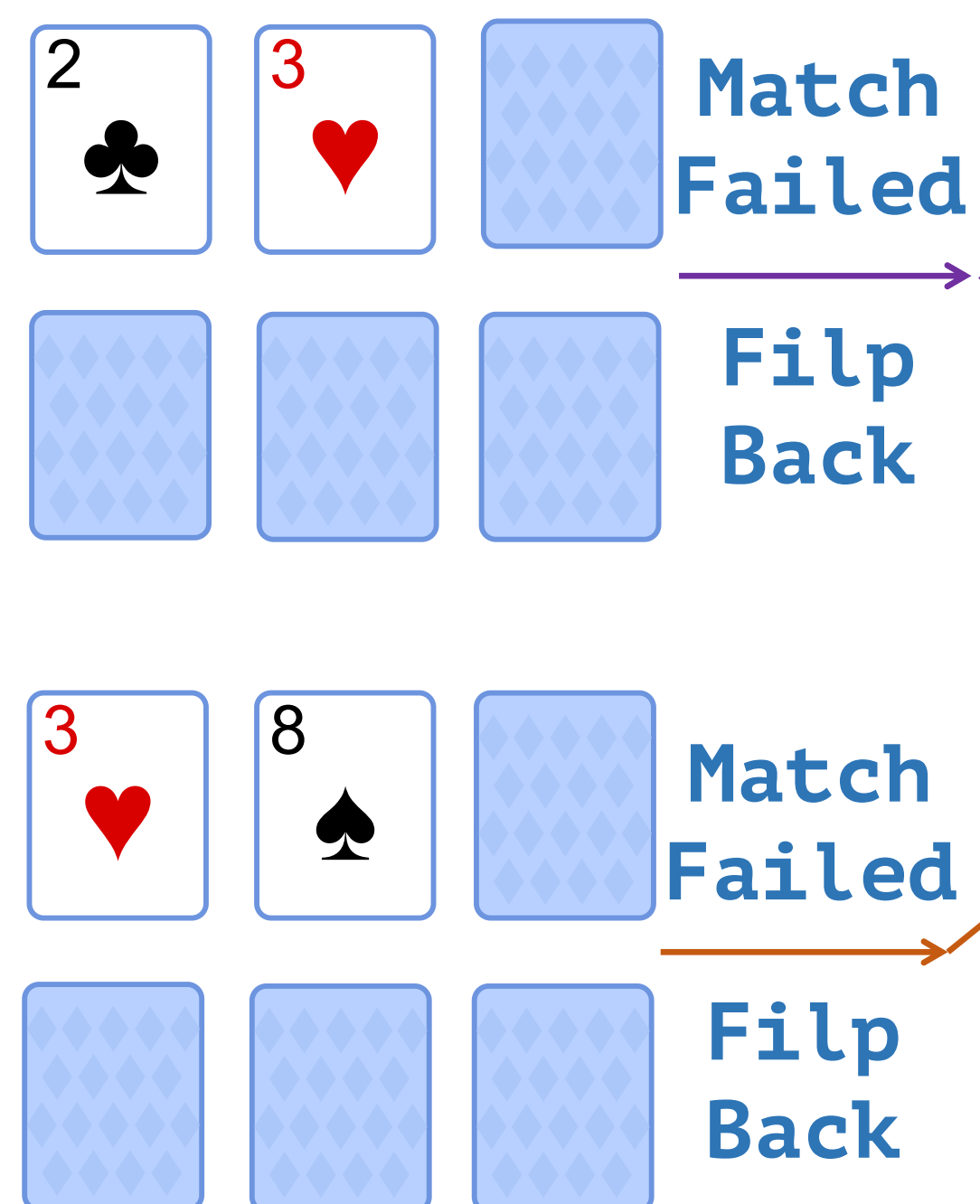
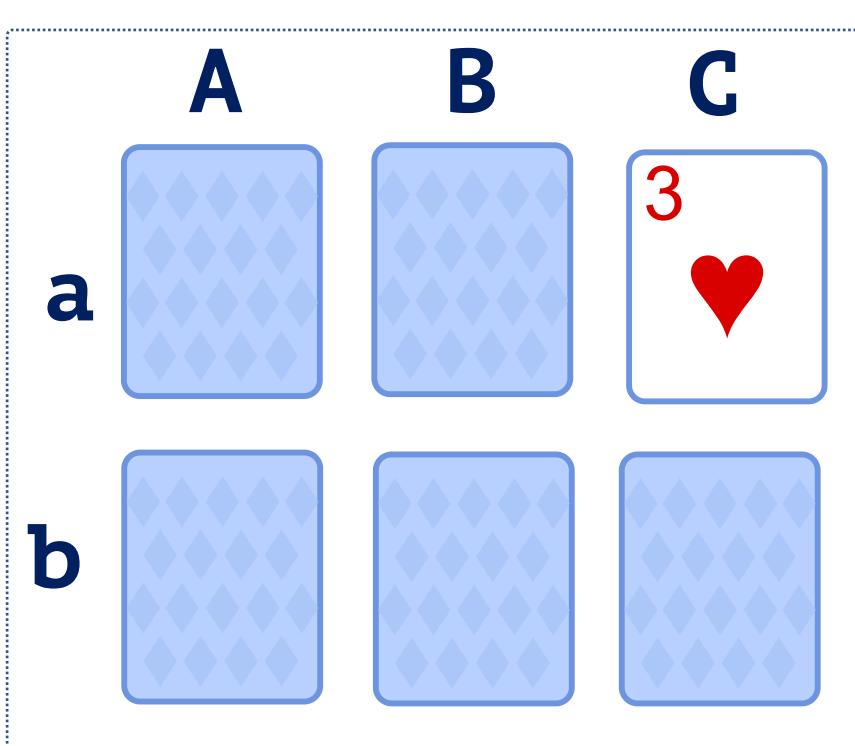
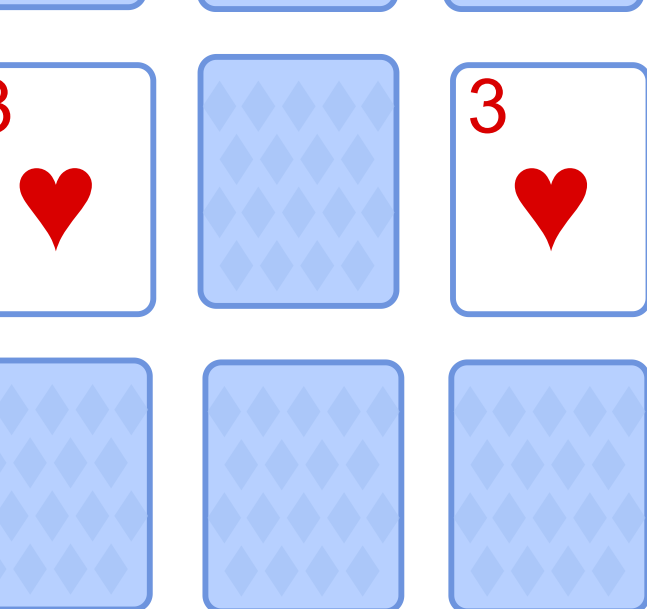
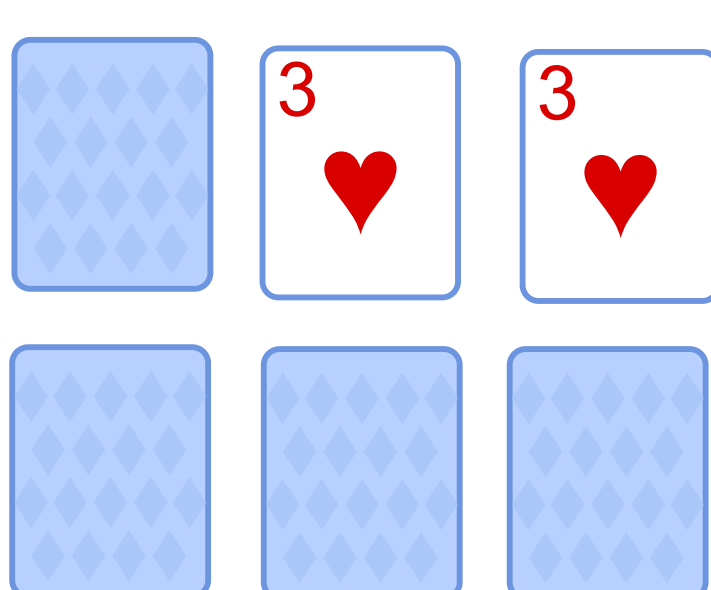


(a) Markov vs. Non-Markov

Markov**Current state is sufficient for Action****Tic-Tac-Toe****Go****Chess**

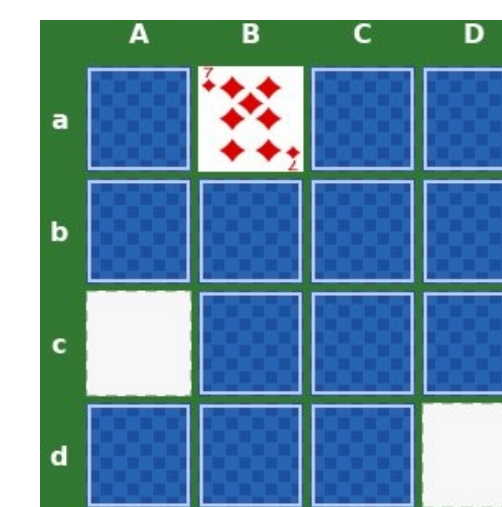
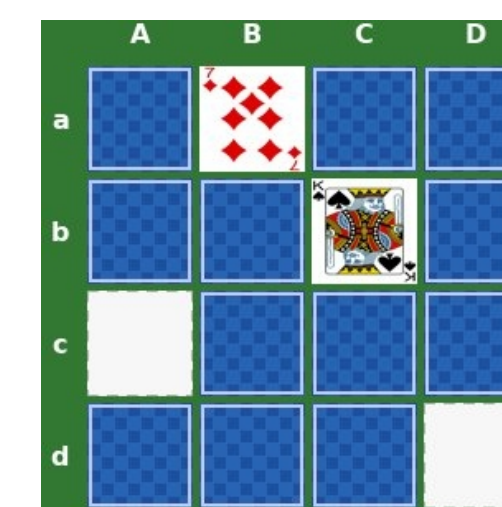
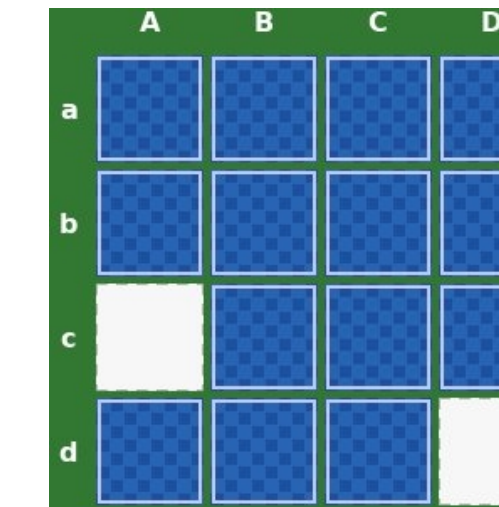
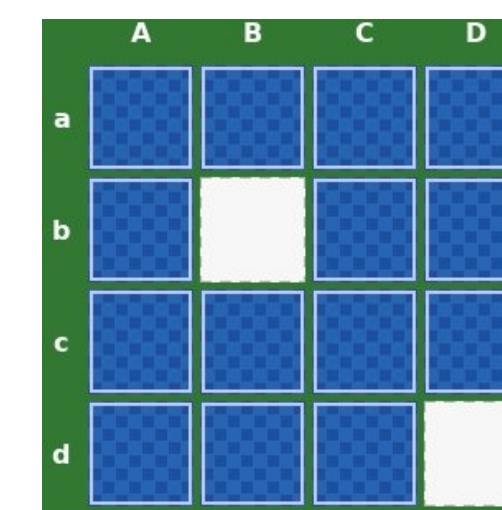
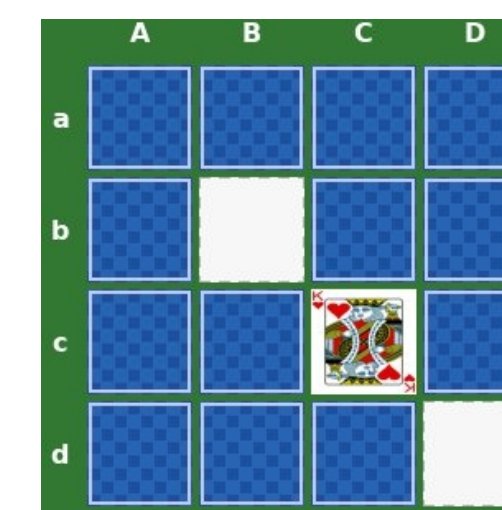
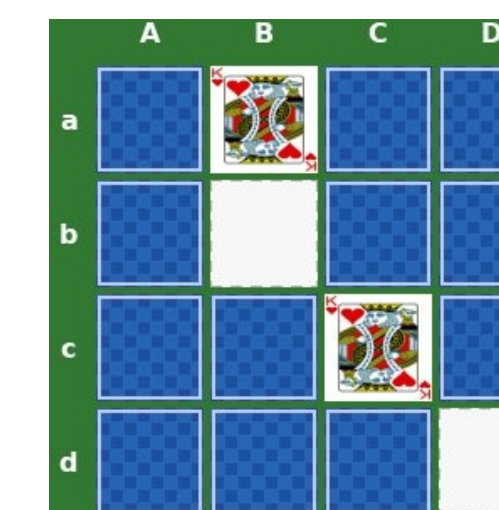
1. Mainly evaluate model reasoning ability
2. Require only current-state understanding

**Non-Markov****Reconstruct state from History first, then Act**Facing the same
Current StateHistory Info is
different
So Action differs**Action: aB****Action: aA**

1. Evaluate memory for action ability
2. Require reconstructing hidden state from history
3. Faulty recall leads to wrong actions



(b) RNG Test Suite

Matching Pairs**State N****State N+1****State N+2****State M****State M+1****State M+2**

**Flip two cards each turn and
remove matched pairs**

Scale

Performance drops
quickly as belief
state size grows

Modality

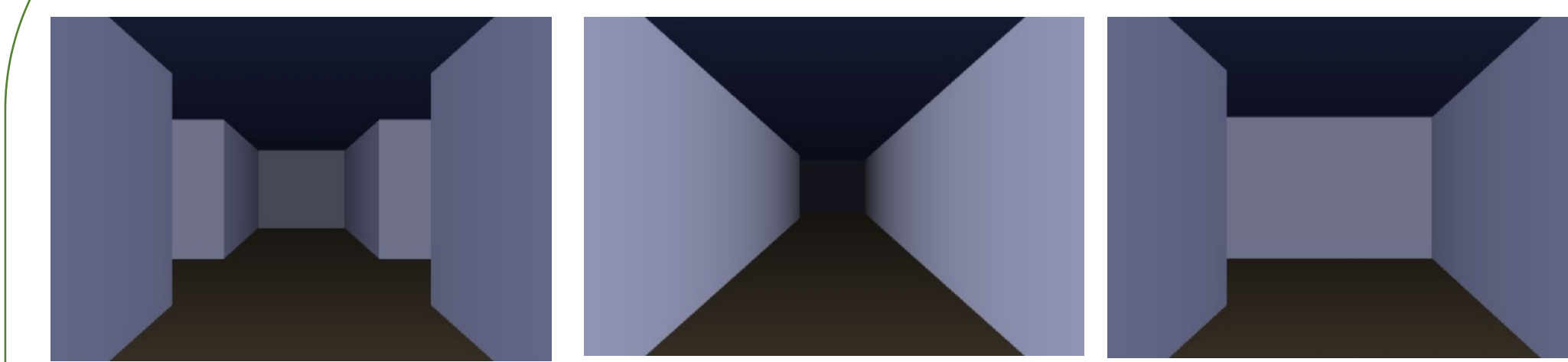
Text works best;
2D/3D visual
tracking is harder

Pattern

State tracking is
sensitive to visual
patterns

MemGap

External memory
helps both, more
strongly for cards

3D Maze**Walking Without Map****Walking With a Minimap**

**Navigate an egocentric 3D
maze from local views**